

AVL Tree Rotations & Operations Data Structures

> **Definition:** An **AVL Tree** (Adelson-Velsky and Landis) is a **self-balancing Binary Search Tree** in which the **Balance Factor (BF)** of every node is either **-1, 0, or +1**.

1. Detailed Technical Explanation

Balance Factor Formula:

Balance Factor (BF) of a node is calculated as:

$$BF = \text{Height of Left Subtree} - \text{Height of Right Subtree}$$

2. The Four AVL Tree Rotations

1. LL Rotation (Single Right Rotation)

- **Cause:** Insertion into the **Left subtree of the Left child** of node z (BF of $z = +2$, BF of left child = +1).

Diagram illustrating LL Rotation:

```
graph TD
    z((z)) --> y((y))
    y --> x((x))
    y --> z
    x --> z
```

2. RR Rotation (Single Left Rotation)

- **Cause:** Insertion into the **Right subtree of the Right child** of node z (BF of $z = -2$, BF of right child = -1).

Diagram illustrating RR Rotation:

```
graph TD
    z((z)) --> y((y))
    y --> x((x))
    y --> z
    x --> z
```

3. LR Rotation (Double Rotation: Left then Right)

- **Cause:** Insertion into the **Right subtree of the Left child** of node z (BF of $z = +2$, BF of left child = -1).

Diagram illustrating LR Rotation:

```
graph TD
    z((z)) --> y((y))
    y --> x((x))
    y --> z
    x --> z
    x --> y
    y --> z
```

4. RL Rotation (Double Rotation: Right then Left)

- **Cause:** Insertion into the **Left subtree of the Right child** of node z (BF of $z = -2$, BF of right child = $+1$).

3. Time Complexity of AVL Tree Operations

| Operation | Average Case | Worst Case |

4. Must-Write Points for Exams

- The height of an AVL tree with N nodes is strictly bounded by $1.44 * \log_2(N)$.

5. Quick Recall Flow
